EC39001: Analog Communication Lab
Experiment 6: Frequency Demodulation

Jai Raj
23EC10031
Group-6

Department of E&ECE
IIT Kharagpur

Date of Experiment: 3rd & 10th *September* 2025

1 Aim of the Experiment

The objectives of this experiment are:

1. To perform **frequency demodulation** of FM signals using different methods.
2. To obtain the demodulated output using **IC 565 PLL**.
3. To design and test a circuit that demodulates the FM signal **without using IC 565**.
4. To analyze FM modulation and demodulation using a **square wave** as the message signal.
5. To demodulate a FM modulated sine wave using previous circuit.

2 Components Used

- IC 565 Phase Locked Loop (PLL)
- Resistors: 330Ω, 560Ω, $10k\Omega$, $47k\Omega$, $4.7k\Omega$, $15k\Omega$, etc.
- Capacitors: $0.001\mu F$, $0.033\mu F$, $0.1\mu F$, $0.03\mu F$, $10nF$, etc.
- Diode
- LM741 Operational Amplifiers
- Potentiometer: $4.7k\Omega$
- Function Generator

- Mixed Signal Oscilloscope (DSO)
- Breadboard and connecting wires
- $\pm 12V$ DC Power Supply

3 Theory

3.1 Frequency Demodulation

Frequency demodulation is the process of extracting the original message signal $m(t)$ from a frequency-modulated (FM) wave. An FM signal is represented as:

$$s(t) = A_c \cos\left(\omega_c t + k_f \int m(t)\, dt\right)$$

where,

- A_c = Carrier amplitude (constant)
- ω_c = Carrier angular frequency
- k_f = Frequency sensitivity
- $m(t)$ = Message signal

The goal of demodulation is to recover $m(t)$ from $s(t)$.

3.2 Demodulation using Phase-Locked Loop (PLL)

A PLL is a feedback system that locks the frequency and phase of a Voltage-Controlled Oscillator (VCO) to an incoming signal.

Main Components:

- **Phase Detector (PD)** – compares the phase of input FM signal and VCO output.
- **Low Pass Filter (LPF)** – removes high frequency components, providing a smooth error voltage.
- **Voltage Controlled Oscillator (VCO)** – generates a frequency dependent on the control voltage.

Demodulation Process:

1. The VCO output tracks the instantaneous frequency of the FM signal.
2. The PD produces a voltage proportional to the frequency deviation.
3. This voltage becomes the demodulated output, proportional to the original message signal $m(t)$.

3.3 Demodulation using Differentiator and Envelope Detector

The instantaneous frequency of an FM wave is proportional to the derivative of its phase:

$$\frac{d}{dt}s(t) = -A_c \sin\left(\omega_c t + k_f \int m(t)dt\right) \cdot (\omega_c + k_f m(t))$$

The amplitude of this differentiated signal is proportional to $m(t)$.

Hence, by differentiating the FM signal and then passing it through:

1. an **envelope detector** (to extract the envelope),
2. and then a **low-pass filter** (to remove high-frequency components),

we obtain the demodulated message signal.

4 Circuit Diagram

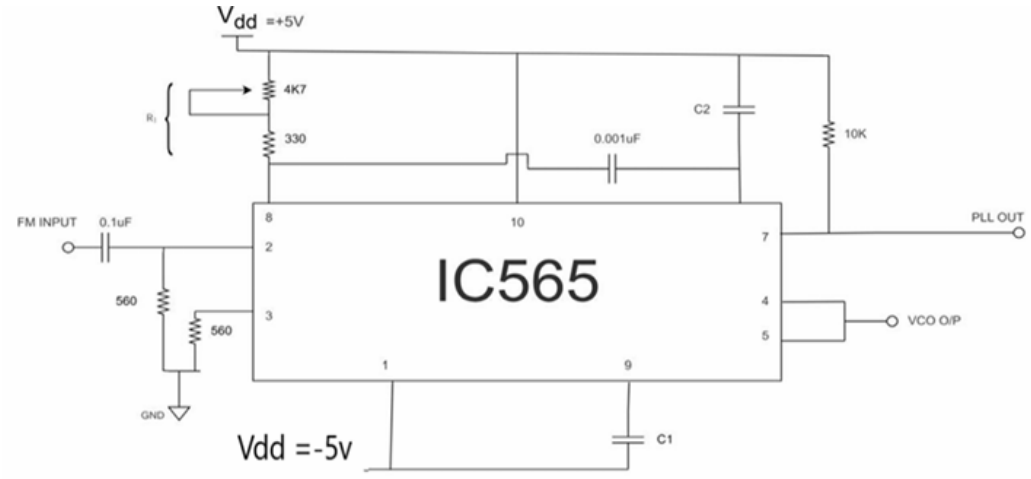


Figure 1: Circuit Diagram for Demodulation using IC 565 (PLL)

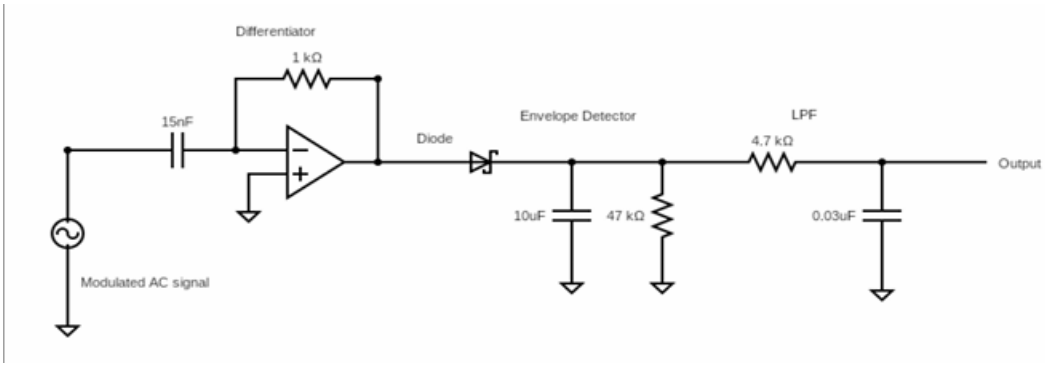


Figure 2: Circuit Diagram for Demodulation without IC 565 (Differentiator + Envelope Detector + LPF)

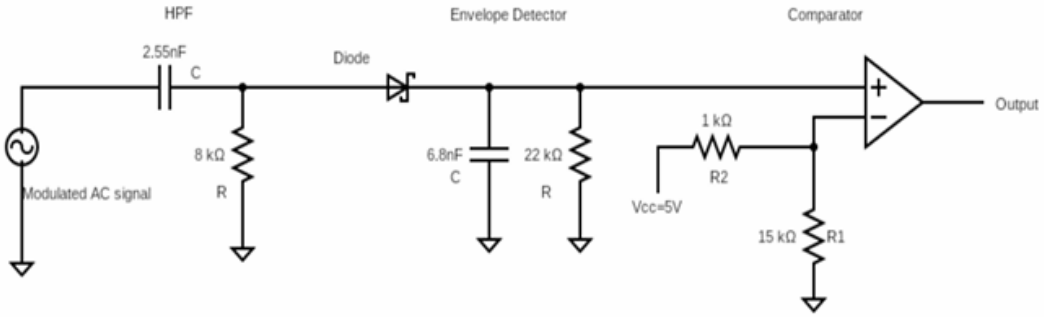


Figure 3: FM Modulation using Square Wave as Message Signal

5 Calculations

- For Task 1: $R_1 = 3.6k\Omega$, $f_o = 10kHz$
  $\Rightarrow C_1 = 6.9nF$
- For Task 2: Envelope Detector design
  $R = 47k\Omega$, $C = 10nF$
- For Task 2: Low Pass Filter design for $f_c \approx 1kHz$
  $R = 4.7k\Omega$, $C = 0.03\mu F$
- For Task 4: High Pass Filter with cutoff near 7.78kHz (to pass 15kHz and block 4kHz)
- Comparator reference setting: For $V_{in} = 5V$, reference voltage $V_{ref} \approx 0.3V$ using $R_1 = 15k\Omega$, $R_2 = 1k\Omega$.

6 Observations and Results

6.1 Task 1: Demodulation using IC 565

We have used the output of the modulator circuit used in last experiment as the input of our demodulator.

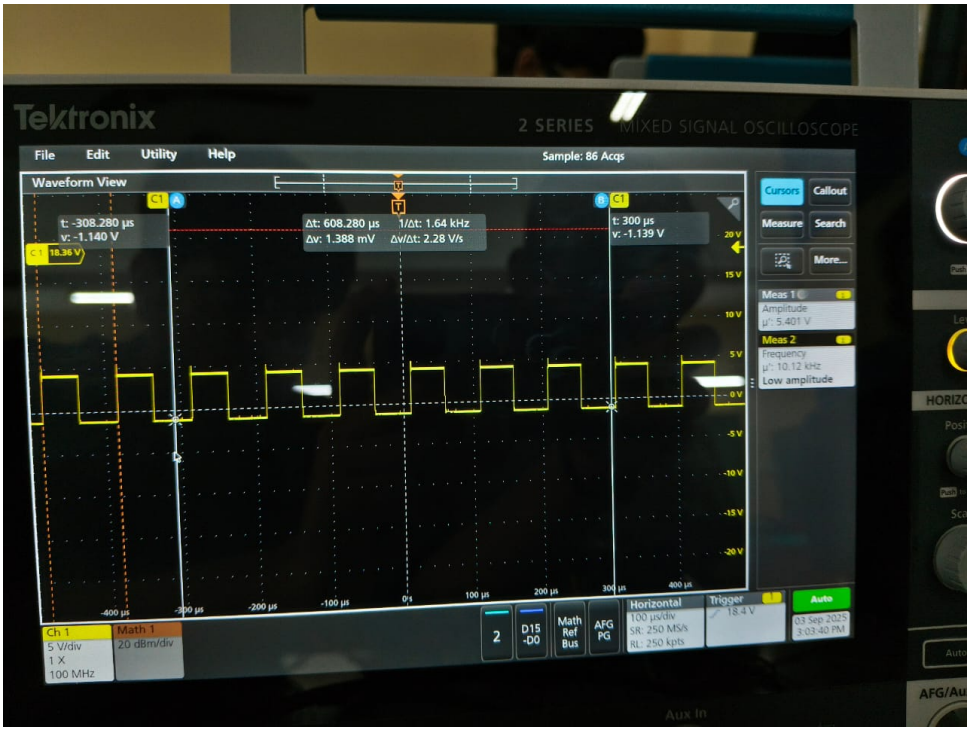


Figure 4: VCO Output (10kHz, 5.4V)

Fig: 4 shows the output of VCO adjusted to 10 kHz in absence of any input signal.

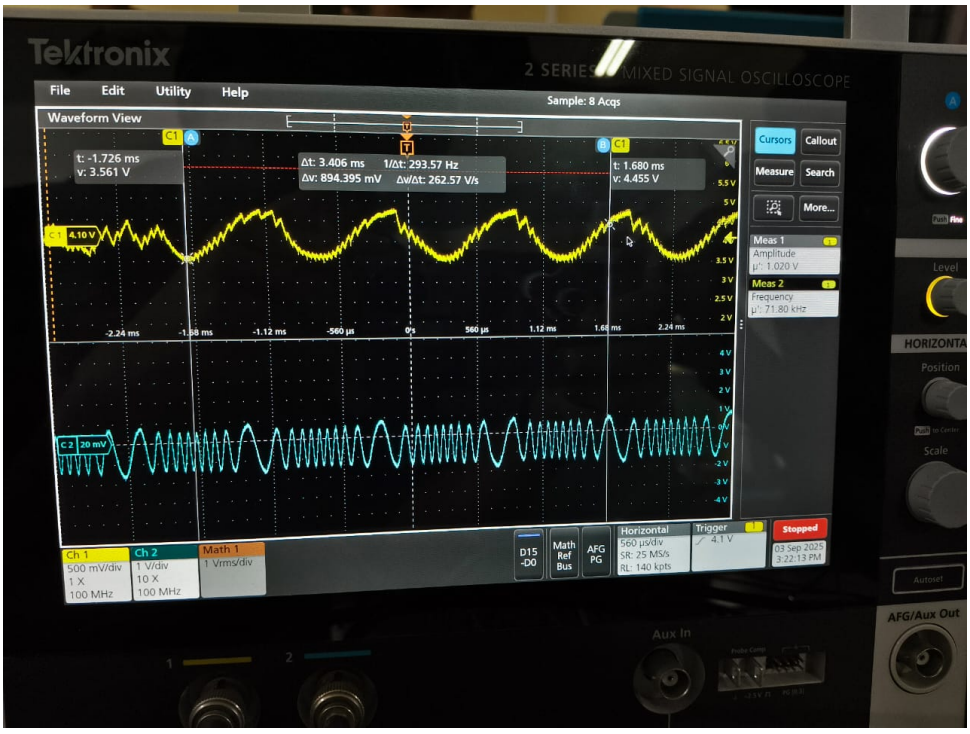


Figure 5: PLL Output: Demodulated (Yellow) and Modulated (Blue) Signal

The PLL successfully demodulated the FM signal.We can confirm by seing the time period of the signal is $1ms$ suggesting 1 kHz signal which is roughly a sine wave. Though higher frequency noise components indicate the need for better LPF filtering.

6.2 Task 2: Demodulation without IC 565

In this part we have tried to demodulate the sine modulated signal as in previous part but without using the pre-built IC instead we built our own circuit to Demoudlute the signal.

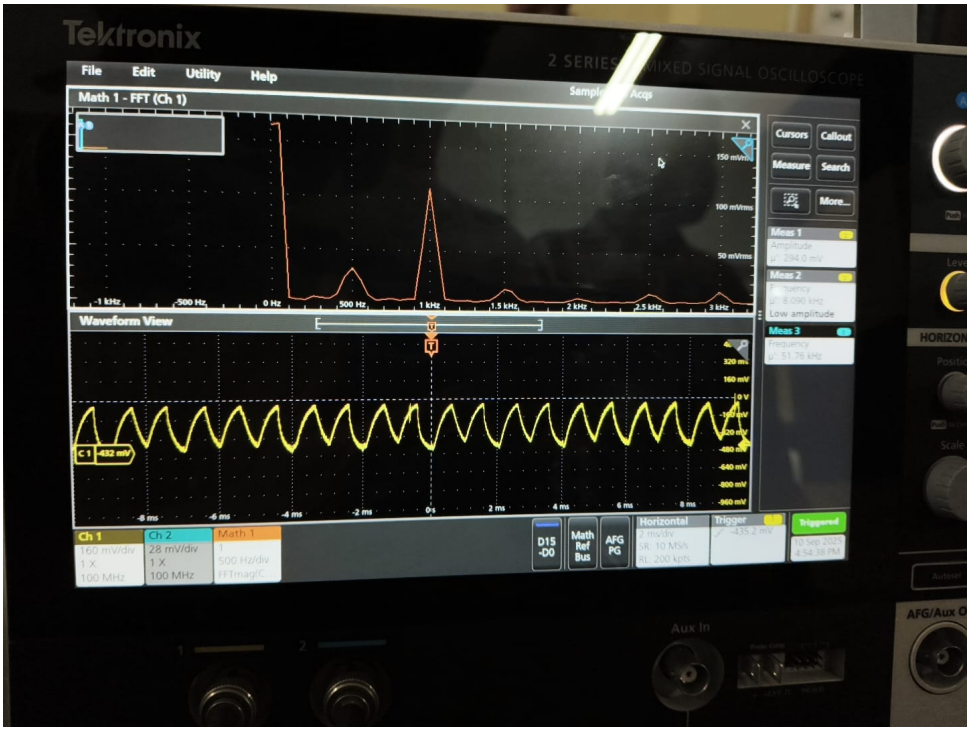


Figure 6: Low-Pass Filter Output — Recovered Message Signal

We obtained a clean demodulated waveform of frequency $2kHz$.But we can observe some distortion like sharp edges and uneven rise and fall waveform.

We can observe several peaks though smaller in amplitude at multiple of 500 Hz.This indicate the cause of distortion in the signal.

Improving the RC values and using a tighter cutoff filter can further refine it.

6.3 Task 3 & 4: FM Modulation and Demodulation with Square Wave as Message Signal

In this part we have modulated(FM) the carrier signal using square wave of 1Khz. and demodulated it using the our own built circuit.



Figure 7: FM Modulated Signal using Square Wave as Message

In Figure 7 we can see the moudulated signal lookes like cut-paste of two sine signals of different frequencies.

We can also observe the frequency spectrum of the modulated wave. It is symmetric around 9.8 kHz not 10 Khz due to drift of carrier wave frequency from originaly set frequency.We can observe several peaks due to odd harmonics with highest peaks around 16 and 4 kHz frequency.

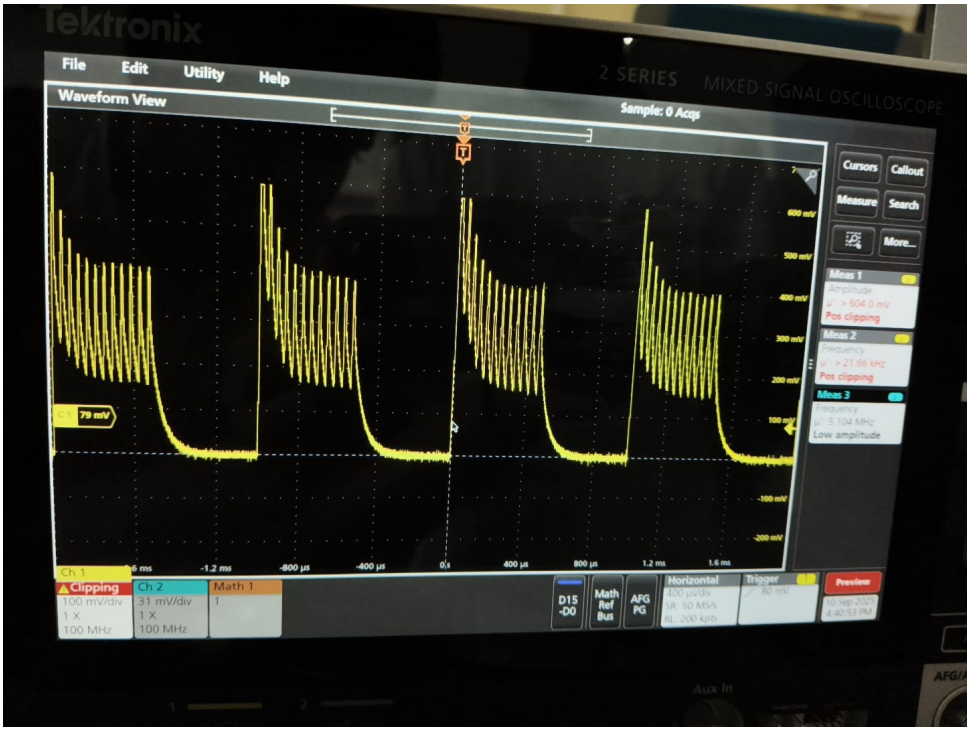


Figure 8: Output of Envelope Detector for Square Wave Demodulation

Figure 8 shows the output of envelope detector when the modulated and rectified signal is passed through it. The envelope follows the sinusoidal envelope of higher frequency components. RC value was tuned to balance charge and discharge time.

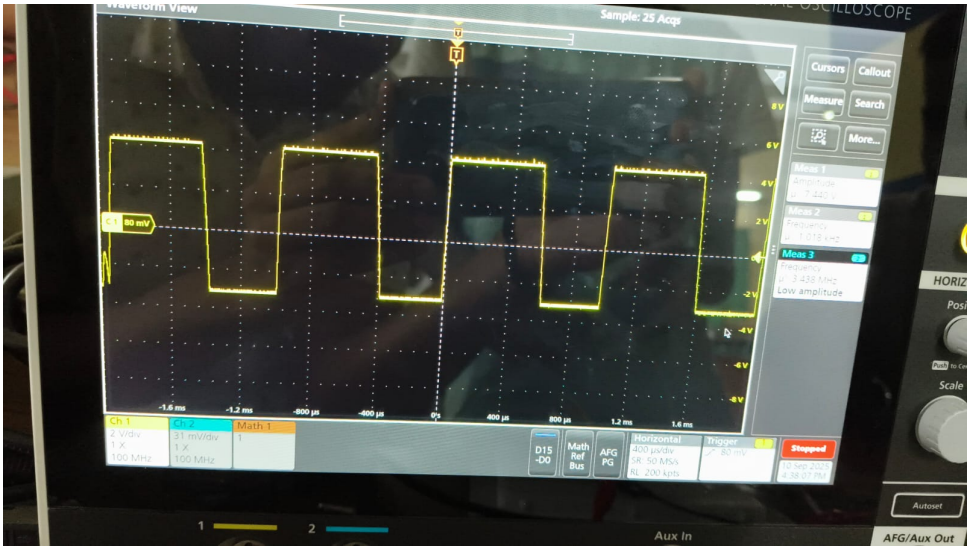


Figure 9: Output of Comparator — Recovered Square Wave

Figure 9 shows the output of Comparator with V_{ref}=0.1 V.We can observe that output waveform is a square wave of frequency 1 kHz .But it's duty cycyle is not 50 % but it's is around 60 % . And have DC offset of around 0.5 V.

Discussion

- The VCO output was observed to be a square wave of frequency around 10 kHz and amplitude approximately 5 V, which matched the expected free-running frequency.
- The PLL successfully demodulated the FM signal, as confirmed by the recovered message frequency of 1 kHz. This verifies that the calculated values of R_1 and C_1 were appropriate.
- However, higher frequency components and noise were visible in the demodulated output, indicating that the internal low-pass filter (LPF) of the PLL was not fully effective in suppressing high-frequency terms.
- For a cleaner output, a higher-order or tighter cutoff LPF can be used either externally or as a replacement for the existing filter.
- In the demodulation without IC 565, a differentiator followed by an envelope detector and LPF was used. The final recovered signal had a frequency of 2 kHz but showed distortion due to improper RC selection.
- The distortion observed was linked to residual high-frequency components around multiples of 500 Hz, which can be reduced by optimizing RC values and the LPF cutoff.
- During FM modulation with a square wave message, the spectrum showed odd harmonics around the carrier (4 kHz, 14 kHz, etc.), consistent with square wave frequency content.
- The envelope detector successfully traced the higher-frequency envelope, and the comparator stage produced a 1 kHz square wave with a duty cycle of about 60% and a small DC offset.
- Minor errors in duty cycle and offset can be corrected by fine-tuning the reference voltage of the comparator and the RC values in the envelope detector.

——END OF REPORT——